

Evaluating Piecewise and Absolute-Value Functions

Answer Key

Algebra 1

Quick Answers

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|--|---|
| 1. $f(0) = 5$, $f(3) = 2$, $f(7) = 6$ | 7. $-4, 2$ |
| 2. -1 | 8. $p(-1) = 3$, $p(-3) = 1$, $p(2) = -2$ |
| 3. $f(0) = 1$, $f(2) = 1$, $f(5) = 22$, — | 9. (a) solving the first rule = 4, (b) solving the second rule = $-4, 4$, — |
| 4. $-2, 8$ | 10. (a) both intersection points = $x = -1$, $y = 0$, $x = 2$, $y = 3$, — |
| 5. $E(74) = 6$, $E(63) = 5$ | |
| 6. $C(2) = 8$, $C(3) = 10$, $C(6) = 25$ | |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-CED.A.1 – *problems 5, 6*

HSA-REI.B.3 – *problems 4, 7*

HSA-REI.C.7 – *problem 10*

HSF-IF.A – *problems 1, 2, 3, 8, 9*

Difficulty 1–5 of 5 across 23 tagged checks.

Common wrong answers

4. *If they got 8: dropped the absolute-value bars and solved $x - 3 + 2 = 7$ only*
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1. Complete the table for $f(x) = |x - 3| + 2$ at $x = 0, 3, 7$.

Solution: Work inside the bars first, then take the distance from zero, then add 2. The value inside can be negative; the value after the bars never is.

$$f(0) = |0 - 3| + 2 = 3 + 2$$

$$f(3) = |3 - 3| + 2 = 0 + 2$$

$$f(7) = |7 - 3| + 2 = 4 + 2$$

$$f(0) = 5, \quad f(3) = 2, \quad f(7) = 6$$

The smallest value is at $x = 3$, where the bars close on zero — that is the corner of the graph.

2. For $g(x) = -2|x + 1| + 5$, find $g(-4)$.

Solution: Inside first: $-4 + 1 = -3$, so $|-3| = 3$. Then multiply by -2 , then add 5.

$$g(-4) = -2(3) + 5 = -6 + 5$$

-1

3. $f(x) = 3x + 1$ for $x < 2$ and $x^2 - 3$ for $x \geq 2$. (a) $f(0)$, $f(2)$, $f(5)$. (b) Why does $x = 2$ use the second rule?

Solution (a): Check the condition before computing. $0 < 2$, so use $3x + 1$. Both 2 and 5 satisfy $x \geq 2$, so use $x^2 - 3$.

$$\begin{aligned} f(0) &= 3(0) + 1 \\ f(2) &= 2^2 - 3 = 4 - 3 \\ f(5) &= 5^2 - 3 = 25 - 3 \end{aligned}$$

$$f(0) = 1, \quad f(2) = 1, \quad f(5) = 22$$

(b) — **instructor-judged.** The second rule carries the condition $x \geq 2$, which *includes* 2; the first stops strictly below 2. Full credit requires that comparison of the two inequality signs. “2 is bigger, so use the second rule” is not the reason — it would give the same answer here and the wrong one on a function written the other way round.

4. For $f(x) = |x - 3| + 2$, find every x with $f(x) = 7$.

Solution: Isolate the absolute value first, then remember that *two* numbers have the same distance from zero.

$$\begin{aligned} |x - 3| + 2 &= 7 \\ |x - 3| &= 5 \\ x - 3 &= 5 \quad \text{or} \quad x - 3 = -5 \end{aligned}$$

$$x = -2 \quad \text{and} \quad x = 8$$

Solving only $x - 3 = 5$ gives 8 and loses the other solution; both are the same distance, 5, from the corner at $x = 3$.

5. $E(t) = |t - 68|$. (a) $E(74)$. (b) $E(63)$.

Solution: The bars turn a signed difference into a distance, which is why a temperature above the setting and one below both give a positive error.

$$\begin{aligned} E(74) &= |74 - 68| = |6| \\ E(63) &= |63 - 68| = |-5| \end{aligned}$$

$$E(74) = 6, \quad E(63) = 5$$

Both answers are in degrees Fahrenheit, and both are positive: an error of five below the setting is still an error of five.

6. $C(h) = 4 + 2h$ for $0 \leq h \leq 3$, and $10 + 5(h - 3)$ for $h > 3$. Find $C(2)$, $C(3)$, $C(6)$.

Solution: 2 and 3 both satisfy $0 \leq h \leq 3$, so the first rule applies to each — note that 3 is included by the $\leq$ sign. Only 6 falls into the second rule.

$$C(2) = 4 + 2(2)$$

$$C(3) = 4 + 2(3)$$

$$C(6) = 10 + 5(6 - 3) = 10 + 15$$

$$\boxed{C(2) = 8, \quad C(3) = 10, \quad C(6) = 25}$$

The two rules agree at $h = 3$, which is what makes this a sensible price list: the cost does not jump at the boundary.

7. For $g(x) = -2|x + 1| + 5$, find every x with $g(x) = -1$.

Solution: Undo the outside operations first, and divide by -2 before opening the bars.

$$-2|x + 1| + 5 = -1$$

$$-2|x + 1| = -6$$

$$|x + 1| = 3$$

$$x + 1 = 3 \quad \text{or} \quad x + 1 = -3$$

$$\boxed{x = -4 \quad \text{and} \quad x = 2}$$

8. $p(x) = x + 4$ for $x \leq -1$ and $2 - x^2$ for $x > -1$. Find $p(-1)$, $p(-3)$, $p(2)$.

Solution: The boundary -1 belongs to the *first* rule this time, because that is where the $\leq$ sits.

$$p(-1) = -1 + 4$$

$$p(-3) = -3 + 4$$

$$p(2) = 2 - 2^2 = 2 - 4$$

$$\boxed{p(-1) = 3, \quad p(-3) = 1, \quad p(2) = -2}$$

9. Using f from Problem 3: (a) solve $3x + 1 = 13$; (b) solve $x^2 - 3 = 13$; (c) which values really solve $f(x) = 13$?

Solution (a): $3x = 12$, so

$$\boxed{x = 4}$$

Solution (b): $x^2 = 16$, so $x = \pm 4$.

$$\boxed{x = -4 \quad \text{and} \quad x = 4}$$

(c) — **instructor-judged.** Only $x = 4$ solves $f(x) = 13$, and the reasoning must run through the conditions, not the arithmetic. The 4 from part (a) is discarded because the rule $3x + 1$ only applies when $x < 2$, and 4 is not. The -4 from part (b) is discarded because the rule $x^2 - 3$ only applies when $x \geq 2$. The 4 from part (b) survives both tests, so it is the

single solution. Keeping a value without testing it against its condition, or rejecting -4 as bad arithmetic, is not full credit.

10. Synthesis challenge. (a) Solve the system $y = x + 1$, $y = x^2 - 1$. (b) What do the solutions mean?

Solution (a): Both rules give y , so set them equal — that is exactly the question “for which inputs do the two models agree?”

$$\begin{aligned}x + 1 &= x^2 - 1 \\0 &= x^2 - x - 2 \\0 &= (x - 2)(x + 1)\end{aligned}$$

So $x = 2$ or $x = -1$, and substituting back into $y = x + 1$ gives $y = 3$ and $y = 0$.

$$\boxed{(-1, 0) \quad \text{and} \quad (2, 3)}$$

(b) — instructor-judged. Each solution is an input at which the two models produce the same output; on a graph they are the two points where the line crosses the parabola. Full credit requires noticing that there are *two* of them — a student who tested a single input and found agreement would have concluded far too much. Restating the coordinates without saying what they mean is not full credit.